Chapter 1

Introduction

Data Executive Summary

This chapter presents the data used to develop and evaluate the closed-pool synonym-ranking system. First, the origin and documented provenance of the source dataset are examined. Subsequently, the composition of the project-specific dataset and the construction of the candidate vocabulary are described. The preprocessing procedure, the train–validation–test partitioning protocol, and the controls introduced to prevent information leakage are then analyzed. Finally, the evaluation subsets, reproducibility requirements, limitations, and ethical considerations are discussed.

The experiments were based on the public TU-Expert-Collection-Topic-Synonyms dataset hosted on Hugging Face. The currently published repository contains 1,266 records and includes English and Dutch terms, English and Dutch synonyms, a topic identifier, and an additional numeric index. However, the repository does not provide a descriptive dataset card explaining the original collection process, annotation methodology, quality-control procedure, or relationship with an upstream authoritative resource. The repository is labeled with an Apache-2.0 license, but its upstream provenance remains unspecified.

For this project, only the English input term, represented by en, and its expected synonym, represented by en_synonym, were retained. The final experimental dataset contains 969 usable rows. One additional row, modernisation–modernisation, was explicitly removed because the input and expected output were identical and the retrieval implementation deliberately excludes the input term from its own candidate list. Ten input terms occur twice with different expected synonyms; consequently, the split was performed by input-term group rather than by individual row.

The task is formulated as closed-pool ranking rather than unrestricted synonym generation. A global candidate vocabulary is constructed from the complete set of retained en_synonym values. For each input term, the system ranks entries from this vocabulary and is evaluated according to the position of the row-specific gold synonym. The candidate vocabulary is fixed and row-independent, and the gold value of the current row is not accessed before prediction. Nevertheless, because labels from validation and test rows are included in the global vocabulary, the protocol is transductive and closed-world. Therefore, the reported results measure ranking quality within a known terminology inventory, not the ability to produce previously unseen synonyms.

A deterministic grouped partition was generated with random seed 42. The resulting data consists of 581 training rows, 194 validation rows, and 194 test rows. The validation set was used for model selection and configuration decisions, while the test set was reserved

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for confirmation of configurations selected beforehand. The training set was not used to fit the reported systems, and the combined 969-row dataset was used only for descriptive reporting and error analysis after the final configuration had been selected.

Several important documentation gaps remain. In particular, the exact filtering rules that reduced the currently published 1,266-row source to 969 usable rows are not fully specified. The final number of unique candidate synonyms, the precise string-normalization procedure, the exact revision of the downloaded source dataset, and the formal definitions of the heuristic term-type categories are also unspecified. These details should be resolved before the thesis is finalized because they affect reproducibility and the interpretation of the evaluation results. Dataset Origin and Scope Public source

The source dataset is published on Hugging Face under the identifier jensjorisdecorte/TU-Expert-Collection-Topic-Synonyms. At the time of review, the repository exposes one split named train, containing 1,266 records. Each record contains the following fields:

Unnamed: 0, an integer index; topic, a string identifier such as topic-12722; en, an English academic or domain term; nl, a corresponding Dutch term; en_synonym, an English synonym; nl_synonym, a Dutch synonym.

The repository is categorized as a text dataset in CSV format and is labeled with the Apache-2.0 license. However, its README or dataset card does not contain a substantive description. Consequently, the public source does not document who created the synonym pairs, how the pairs were annotated, whether domain experts validated them, which domains are represented, or how disagreements were resolved.

The name of the dataset suggests a connection with a Tilburg University expert-topic collection. However, no authoritative upstream citation, Digital Object Identifier, version history, or data-generation description is provided by the Hugging Face repository. For this reason, such a connection must not be presented as confirmed solely from the repository name. In the thesis, the Hugging Face repository can be cited as the immediate data source, while the original institutional and annotation provenance should be marked as unspecified unless it can be confirmed by the dataset publisher.

The public preview shows that the source is bilingual and contains heterogeneous terminology. Examples cover fields such as economics, law, psychology, information systems, religion, education, and public policy. The source also contains records in which en is a hyphen or a null value, while en_synonym remains populated. This observation is relevant because the final project dataset contains substantially fewer rows than the public source, indicating that additional filtering took place. Project-specific extraction

The project reduced the source data to the two columns required by the ranking task:

en, which represents the input query term; en_synonym, which represents the row-specific relevant candidate.

The topic, nl, and nl_synonym fields were not passed to the evaluated methods. This decision simplified the task to English-only synonym ranking and prevented auxiliary fields from influencing candidate generation or ranking. In particular, the experiment log states that the preparation script retains only en and en_synonym, and that topic does not reach the downstream methods.

Table 1 compares the currently published source repository with the final project dataset. Characteristic Public repository Project dataset Documentation status Total rows 1,266 969 Both counts documented Published splits One train split Train, validation, test, and derived full set Documented English input field en en Documented English synonym field en_synonym en_synonym Documented Dutch fields nl, nl_synonym Removed Documented

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Topic identifier Present Removed before modeling Documented Numeric source index Present Use in final data unspecified Partially documented Source-data revision Repository is public Revision or commit not pinned Unspecified Final unique input terms Not reported 959, inferred from 969 rows and ten duplicated inputs Inferred Final unique candidate synonyms Not reported Unspecified Must be recomputed License Apache-2.0 label Reused under source label Documented at repository level Original annotation process Not described Not independently documented Unspecified

The source-to-project reduction requires further clarification. Relative to the currently published 1,266-row repository, 297 rows do not appear in the final 969-row evaluation dataset. The supplied project documentation explicitly accounts for only one excluded record: the identity pair modernisation–modernisation. It does not provide a complete rule explaining the other exclusions, such as whether rows with null inputs, hyphen placeholders, empty strings, missing synonyms, duplicated pairs, or invalid encodings were removed.

This discrepancy does not necessarily indicate an implementation error. For example, the project may have used a different source revision or may have applied legitimate validity filters. However, the transformation cannot currently be reproduced from the thesis materials alone. Therefore, the data chapter should include either the exact filtering code or an explicit sequence of conditions, together with row counts after each condition. Unit of observation and relevance labels

Each retained row represents one evaluation instance. The input is one academic or domain term, and the row-specific target is one expected English synonym. Consequently, the dataset is not a conventional multiclass classification dataset with a small set of balanced classes. Instead, it is a ranking dataset in which each query has one labeled relevant item within a much larger shared candidate vocabulary.

For a candidate vocabulary of size (V), each row contains one labeled positive candidate and (V-1) candidates treated as negative for that particular row. The precise value of (V) is currently unspecified because the final number of distinct en_synonym strings is not reported. An early experiment recorded 921 unique candidate values for an earlier dataset state, but that historical value should not be reused as the final vocabulary size without recomputing it from the final 969-row dataset.

Ten input terms occur twice and are associated with two different gold synonyms. Given 969 rows and ten terms occurring twice, the project dataset contains 959 distinct input strings, assuming each documented duplicate occurs exactly twice. This is an arithmetic inference from the supplied description rather than a separately reported statistic.

The repeated inputs also expose a limitation of single-label evaluation. If a duplicated term has two valid synonyms, each row recognizes only one of them as correct. A system may therefore return the alternative synonym associated with the other row and still be scored as incorrect. More generally, the dataset can contain plausible synonyms that are not recorded as the current row's gold value. The project's error analysis provides several examples in which the top-ranked output is semantically defensible but differs from the single annotated target. Dataset Composition and Candidate Space Term-type distribution

The full 969-row dataset was divided during error analysis into four mutually exhaustive heuristic term types: abbreviation-like terms, compounds, multi-word phrases, and single words. The resulting counts sum to the full dataset size. However, the exact regular expressions, tokenization rules, and category-precedence rules used by the classification script are not specified in the narrative documentation. Therefore, these labels should be described as project-specific analytical categories rather than authoritative linguistic classes.

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Table 2 summarizes the available distribution. The two failure-rate columns refer to the final full-dataset evaluation of the selected system. A ranking error means that the expected synonym was retrieved within the top five but was not ranked first. A retrieval miss means that the expected synonym was absent from the top five and could not be recovered by the reranking stage. Term type Rows Dataset share Ranking-error rate Retrieval-miss rate Abbreviation-like 34 3.5% 5.9% 8.8% Compound 432 44.6% 11.3% 3.2% Multi-word phrase 250 25.8% 12.8% 4.4% Single word 253 26.1% 14.2% 9.9% Total 969 100.0% — —

The distribution shows that compounds form the largest category, accounting for approximately 44.6% of the dataset. Multi-word phrases and single words each represent approximately one quarter, while abbreviation-like inputs form a comparatively small subset. Under the tested conditions, single-word terms were the most difficult category for both ranking and retrieval. Their retrieval-miss rate of 9.9% was more than twice the corresponding rate for compounds and multi-word phrases.

These rates should not be interpreted as intrinsic properties of single words in general. The categories are not balanced, their definitions are heuristic, and no confidence intervals are reported. In addition, a single word is often more polysemous than a longer technical expression because less lexical context is available to identify the intended sense. This explanation is consistent with the observed error pattern, but it remains an interpretation rather than a causal result established by the experiment. Domain and frequency distributions

A formal distribution by academic domain is not available. Although the source provides a topic field, the values shown in the public repository are identifiers such as topic-12722, rather than documented subject categories. Furthermore, the project deliberately removes the field before modeling. Consequently, no reliable grouping into economics, law, computing, psychology, or other disciplines can be reconstructed from the supplied files without an additional manual or external classification step.

A frequency-based distribution is also unavailable. Neither the public dataset nor the retained project table contains corpus frequency, document frequency, query frequency, or usage counts. Therefore, the terms cannot presently be divided into high-frequency and low-frequency subsets using the dataset alone. Any future frequency analysis would require a defined external corpus, a documented matching procedure, and a fixed corpus version. Otherwise, the meaning of “frequent” would remain dependent on the selected external source.

The absence of explicit domain and frequency metadata limits the interpretation of aggregate results. For example, a method could perform well because the dataset contains many compounds from semantically coherent domains, while performing less effectively on rare terminology from specialized areas. Aggregate Hit@1 would not reveal this behavior. Therefore, the current term-type analysis is useful, but it does not replace domain- or frequency-stratified evaluation. Definition of the closed candidate pool

The project adopts a closed-pool ranking formulation. Let the final dataset be

$$D = \{(q_i, y_i)\}_{i=1}^{N},$$

where q_i is the English input term, y_i is its recorded English synonym, and $N = 969$. The global candidate vocabulary is defined as

$$C = \mathrm{unique}\left(\{y_i \mid (q_i, y_i) \in D\}\right).$$

For every query q_i, the system produces an ordered list of candidates selected from (C). The raw output of any Large Language Model (LLM) used for query expansion is not treated

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as the final answer unless it corresponds to an entry in this vocabulary. Even the methods described as zero-shot use the generated text as an additional retrieval query and return a vocabulary entry rather than unconstrained text.

The closed-pool formulation provides several engineering advantages. First, every returned value belongs to a controlled terminology inventory. Second, exact-match ranking metrics can be computed without introducing a separate semantic-equivalence evaluator. Third, different retrieval and reranking architectures can be compared over the same candidate space. Finally, the reranker can be constrained to reorder candidates rather than generate unsupported values.

However, the formulation also narrows the meaning of the task. The system is not required to discover a synonym that is absent from the dataset. Instead, it must identify the expected item from a known inventory that already contains that item. Therefore, the experiment evaluates candidate discrimination and ordering under a closed-world assumption. It does not establish performance for open vocabulary expansion, terminology mining from unseen corpora, or synonym generation for terms whose correct alternatives are outside the original inventory.

The project also excludes the input term itself from its retrieval results. This prevents a system from satisfying the task by returning the original query unchanged. However, it creates a structural problem for identity pairs in which $q_i = y_i$. The documented modernisation–modernisation row was removed for this reason: after self-exclusion, the only labeled correct candidate would become unavailable.

Several construction details remain unspecified and should be recorded in the final implementation documentation:

whether deduplication occurs before or after whitespace and Unicode normalization; whether candidate equality is case-sensitive; how null values and empty strings are handled; whether the candidate list is sorted or preserves source order; how ties between candidates with equal scores are resolved; whether duplicated synonym strings retain any source-row identifiers; the final cardinality and cryptographic hash of the candidate vocabulary.

These details can affect exact-match evaluation and tie-breaking. Therefore, they are part of the experimental definition rather than minor implementation details. Preparation and Normalization Documented preparation pipeline

The preparation procedure was designed to produce an English term–synonym table suitable for retrieval experiments. The following operations are explicitly supported by the project documentation:

The source dataset was loaded and reduced to the en and en_synonym columns. Invalid or unusable source rows were removed, although the complete validity criteria are unspecified. The identity pair modernisation–modernisation was excluded. Repeated input terms were retained because they represent different recorded synonyms. The full en_synonym column was deduplicated to form the shared candidate vocabulary. Records were grouped by en before partitioning so that repeated inputs could not cross split boundaries. The same fixed candidate vocabulary was made available to every split.

The resulting logical flow is illustrated in Figure 1. Dashed or explicitly marked steps correspond to operations whose detailed rules are not fully documented.

Public Hugging Face dataset 1,266 rows

Select English fields en and en_synonym

Filter unusable records exact rules unspecified

Remove identity pair modernisation = modernisation

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Final project table 969 rows
Extract and deduplicate all en_synonym values
Fixed global candidate vocabulary final unique count unspecified
Group records by exact en value
Seeded grouped partition seed = 42
Train 581 rows
Validation 194 rows
Test 194 rows
Candidate ranking
Read row-specific gold label only for evaluation
Figure 1. Dataset preparation, candidate-vocabulary construction, and grouped partitioning. The candidate pool is constructed globally and remains constant across splits, while the row-specific target is accessed only after a prediction has been produced. Tokenization and casing

No project-level tokenization procedure is documented. The retained terms appear to be stored and passed as text strings, while tokenization is performed internally by the selected embedding or language model. Therefore, token boundaries are model-dependent rather than defined once during dataset preparation. This distinction is important because the term-type and lexical-overlap analyses use their own heuristic tokenization, which may not match the subword tokenization used by the retrieval models.

No global lowercasing step is documented. The public source contains mixed capitalization, including examples such as Firm relocation, Americanisation, and Antarctic, Arctic, Polar Regions. Consequently, the final thesis should not state that the data was lowercased unless this behavior is verified directly in data/prepare_dataset.py or the exported CSV.

Preserving case can be beneficial for acronyms and proper names. For example, distinctions involving IT, IdM, or named regions can carry semantic information. However, case preservation may also cause logically equivalent variants to remain distinct in exact-string deduplication. Therefore, the project should specify whether case affects candidate identity, self-exclusion, and metric computation. Unicode, diacritics, punctuation, and whitespace

No Unicode normalization form, such as NFC or NFKC, is documented. Similarly, no explicit policy is given for diacritic removal, apostrophe normalization, dash normalization, repeated whitespace, surrounding whitespace, slashes, commas, or parentheses. These details are particularly relevant because the public source includes multilingual text and visibly heterogeneous punctuation.

The Dutch fields, where diacritics are more frequent, are removed before modeling. Nevertheless, this does not eliminate the need for Unicode normalization because English terminology can contain typographic apostrophes, accented proper names, non-ASCII dashes, or encoding artifacts. Moreover, string equality is used in dataset filtering and exact-match evaluation. Two visually similar strings can therefore be treated as different candidates if their underlying Unicode representations differ.

At minimum, the final preparation script should document whether the following transformations are applied: Normalization decision Status in supplied documentation Recommended thesis treatment Trim leading and trailing whitespace Unspecified Verify and report Collapse repeated internal whitespace Unspecified Verify and report Convert empty strings to missing values Unspecified Verify and report Remove hyphen placeholders Implied by source-to-final reduction, not documented State exact rule Remove null en values Implied, not documented State exact rule Unicode normalization Unspecified Record the selected

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form or state none Lowercasing Not documented Do not claim it was performed Diacritic removal Not documented Prefer preservation unless justified Punctuation normalization Unspecified Verify and report Exact duplicate-row removal Unspecified Report count and rule Candidate-string deduplication Required by pool construction Report order and normalization basis Stemming, lemmatization, and morphological processing

Stemming and lemmatization were not adopted as mandatory preprocessing for the selected ranking pipeline. Morphological query variants, including changes related to singular and plural forms, hyphens, spaces, and parenthetical text, were evaluated experimentally. Under the tested conditions, these variants were neutral or slightly detrimental, and the method was discarded.

A stem-sharing indicator was later used as a diagnostic feature during error analysis. However, it did not modify the dataset or the system input. This distinction should be preserved in the chapter: stemming was used to study failure patterns, not to normalize the production data.

The decision not to apply stemming or lemmatization globally is reasonable for this task. Academic terminology often contains meaningful distinctions in number, derivational form, or word composition. Aggressive normalization could merge candidates that should remain separate or reduce the ability to distinguish a synonym from a merely related term. In addition, modern embedding models apply learned subword tokenization and can already represent many morphological relationships without rewriting the source text. Model-specific textual preparation

The intfloat/e5-base-v2 embedding model was used with the query and passage prefixes expected by its retrieval formulation. Input queries receive the query: prefix, while candidate entries receive the corresponding passage: prefix. These prefixes are model-interface requirements and should not be confused with permanent changes to the stored dataset.

This separation is important for reproducibility. Dataset normalization defines the canonical strings used for candidate identity and evaluation, whereas model-specific preprocessing defines how those strings are presented to a particular model. The canonical data should remain unchanged when an embedding model is replaced. Otherwise, a model comparison could unintentionally include differences in dataset preparation. Partitioning, Leakage, and Evaluation Subsets Grouped train–validation–test protocol

The final 969 rows were partitioned using data/make_splits.py. A fixed seed of 42 and a nominal 60/20/20 allocation produced 581 training rows, 194 validation rows, and 194 test rows. The observed proportions are 59.96%, 20.02%, and 20.02%, respectively.

The split was performed by unique en value rather than by row. Therefore, all occurrences of a repeated input term were assigned to the same split. This rule prevents a term with one synonym from appearing in one split while the same input with another synonym appears in another. Without grouping, the system-selection process could benefit from direct exposure to the same query string outside the test set.

Table 4 summarizes the role assigned to each partition. Partition Rows Percentage Intended use Observed project use Train 581 59.96% Parameter fitting or future calibration Reserved; not used by the reported methods Validation 194 20.02% Architecture and hyperparameter selection Used for model, fusion, threshold, and pipeline decisions Test 194 20.02% Final evaluation of selected configurations Used for confirmation after decisions were made Full 969 100.00% Not an independent split Derived union used for descriptive reporting and error analysis

Although no neural model was trained from scratch, the separation remains necessary. Se-

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lecting an embedding model, fusion strategy, query-expansion setting, candidate-pool width, or reranker according to observed performance constitutes data-dependent model selection. Repeatedly making such decisions on the test set would overfit the experimental process even if no gradient-based parameter update were performed.

The training partition was left unused. This choice protects a substantial subset for future fitting or calibration, but it also means that several manually selected parameters were chosen using validation results rather than estimated from training data. For example, a future version could calibrate fusion weights or confidence thresholds on the training set, use validation for model selection, and reserve test strictly for the final estimate.

The documentation states that the test set was accessed once per already-decided configuration. This is more permissive than a single final test evaluation across the entire project because several successively selected configurations were confirmed on test. Consequently, the test set remained protected from direct parameter optimization, but repeated reporting across experimental cycles can still create indirect familiarity with test performance. The final thesis should distinguish the definitive fallback-verified result from earlier exploratory confirmations. Leakage definition

For this project, label leakage is defined narrowly as accessing the current row's en_synonym before producing that row's ranked output. The full set of synonym strings may be used as a fixed candidate vocabulary, provided that no method performs a row-specific lookup of the expected answer. Ground truth is read only after candidate generation and ranking, when the prediction is evaluated.

This definition is internally consistent with a closed-pool task, but several distinct leakage concepts should be separated: Leakage risk Operational definition Implemented control Residual consideration Row-label leakage Reading the current row's gold synonym before prediction Gold is accessed after ranking Requires code-level verification Input-group leakage Same en value occurring across different splits Grouped partition by exact en Near-duplicates and spelling variants are not documented Test-selection leakage Choosing methods or parameters from test scores Decisions made on validation Multiple configuration confirmations were still reported Auxiliary-feature leakage Supplying fields unavailable in the intended use case Only en is passed as query input Exact source index handling is unspecified Full-data tuning leakage Using the 969-row union to choose a method Full set used only after configuration lock Must be maintained in future experiments Candidate-vocabulary exposure Test labels included in global candidate inventory Explicitly allowed by task definition Creates a transductive closed-world setting Identity leakage Input string is also accepted as its synonym Input candidate excluded; identity row removed Equality normalization is unspecified

The most important residual issue is candidate-vocabulary exposure. Because (C) is constructed from all en_synonym values before splitting, every test target is guaranteed to exist in the search inventory. This is not leakage under the project's declared ranking task, because the system is intended to rank a predefined terminology pool. However, it would be leakage under an inductive evaluation intended to measure generalization to unseen synonym labels.

Therefore, the task should be described explicitly as transductive closed-pool synonym ranking. The term transductive indicates that the output vocabulary is known globally at evaluation time, while the correct mapping between each test input and a vocabulary item remains hidden. This wording prevents readers from interpreting the results as open-ended synonym-generation accuracy. Evaluation subsets

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The dataset supports several analytical subsets, although only some have been evaluated quantitatively. Subset dimension Availability Definition status Appropriate use Term type Available Counts available; exact heuristic unspecified Compare single words, phrases, compounds, and abbreviations Lexical overlap Available post hoc Low/high bin thresholds unspecified Analyze surface-distance effects Ambiguous or duplicated input Ten input terms identified Subset metrics not reported Measure sensitivity to multiple valid labels Academic domain Unavailable No documented domain labels retained Requires new annotation or mapping Corpus frequency Unavailable No frequency field or reference corpus Requires an external fixed corpus Input length Derivable No reported bins Analyze token- or character-length effects Retrieval difficulty Derivable post hoc Gold present or absent from top five Distinguish retrieval and ranking failures Gold rank before reranking Available in analysis outputs Documented for the final system Measure causal effect of reranking

The lexical-overlap analysis found that rows with no shared tokens between input and gold had a retrieval-miss rate of 8.9% and a ranking-error rate of 16.0%, compared with 1.7% and 6.9% for the reported high-overlap group. However, the precise tokenization and threshold defining "high overlap" are not provided in the narrative files. These values are therefore informative but not fully reproducible without the associated analysis script.

The project also separates final errors into ranking errors and retrieval misses. On the full 969-row dataset, 797 predictions were correct at rank one, 119 contained the gold synonym within the top five but not at rank one, and 53 did not retrieve the gold within the top five. These categories help identify whether a failure originates before or after reranking. However, they are outcome-defined groups rather than independent pre-experimental subsets and should not be used as if they were fixed strata for unbiased model selection.

Ambiguity deserves a separate analysis because ten input terms have multiple annotated gold synonyms. Suitable reporting would include the number of those rows for which the alternative recorded synonym was returned, as well as a multi-positive evaluation in which either documented synonym is accepted. These results are currently unspecified. Metrics and Reproducibility Ranking metrics

The evaluation module reports Hit@1, Hit@3, Hit@5, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain at cutoffs three and five, written as NDCG@3 and NDCG@5. These measures evaluate complementary properties of the ranked candidate list.

Let r_i denote the rank of the gold synonym for row (i), with $r_i = \infty$ when it is absent from the returned list. Hit@*k* is defined as

$$\mathrm{Hit@}\,k = \frac{1}{N}\sum_{i=1}^{N}\mathbb{1}[r_i \le k].$$

Hit@1 measures exact first-choice accuracy. Hit@3 and Hit@5 measure whether the relevant item is available in a short candidate list. Therefore, the latter metrics are particularly useful for identifying whether errors could be corrected by a stronger reranker or by human review.

MRR is defined as

$$\mathrm{MRR} = \frac{1}{N}\sum_{i=1}^{N}\begin{cases}1/r_i, & r_i < \infty,\\ 0, & r_i = \infty.\end{cases}$$

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MRR gives full credit to a rank-one result and progressively less credit to lower ranks. Consequently, it distinguishes systems that have the same Hit@5 but place the correct item at different average positions.

NDCG is based on cumulative gain discounted according to rank and normalized against the ideal ranking. It was introduced to reward systems that place more relevant results earlier in a ranked list. In the present dataset, each row has one binary gold item. Therefore, assuming the implementation uses standard logarithmic discounting, NDCG@k simplifies to

$$\mathrm{NDCG@}\,k_i = \begin{cases} 1/\log_2(r_i+1), & r_i \le k, \\ 0, & r_i > k. \end{cases}$$

The final score is averaged across rows. Under this single-relevant-item formulation, NDCG and MRR are related but not identical because their discount functions differ.

The metrics map to the dataset and its subsets as follows:

Hit@1 is the primary measure of whether the system selects the row-specific synonym as its final answer. Hit@3 and Hit@5 measure shortlist coverage and are especially relevant to retrieval-stage recall. MRR measures overall sensitivity to the first relevant item's position. NDCG@3 and NDCG@5 provide position-sensitive evaluation within the operational shortlist. Absolute hit counts should accompany proportions, particularly for the 194-row validation and test sets, where one changed prediction corresponds to approximately 0.52 percentage points. Subset metrics should be recomputed independently within each term type or ambiguity group rather than inferred from global results.

Because each row contains only one accepted gold label, the metrics do not reward other valid synonyms. This is a limitation of the annotation scheme rather than of the metric implementation. A future evaluation could use multiple relevance labels per query or graded judgments, but this would require additional annotation. Reproducibility controls

The data partition is reproducible in principle because the script name, grouping rule, split ratio, and seed are documented. The primary regeneration command is:

bash

python data/make_splits.py

The unified evaluation harness is executed through run_eval.py, and each completed run writes aggregate and row-level outputs. The project maintains a master experiment log containing the method, split, row count, metrics, absolute hit counts, latency, configuration note, timestamp, and reranker fallback count. Per-run JSON summaries and detailed CSV files are also stored.

These controls provide useful traceability. In particular, row-level files make it possible to verify whether a reported improvement results from a changed candidate set or merely from nondeterministic reranking. Nevertheless, data reproducibility is not complete because several environmental and versioning details remain unspecified.

A complete reproducibility package should record the following items:

Source revision: the exact Hugging Face dataset commit or revision used for download. Source checksum: a SHA-256 hash of the original downloaded file. Cleaning audit: the number of rows before and after each filtering condition. Canonical schema: column names, data types, missing-value policy, and text encoding. Normalization policy: casing, Unicode, whitespace, punctuation, and placeholder handling. Candidate inventory: final unique candidate count, ordering rule, and checksum. Split manifest: exact row identifiers and input groups assigned to each split. Code revision: a repository commit hash corresponding to

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the reported experiments. Environment specification: Python version and pinned versions of data-processing and model libraries. Randomness controls: seeds for splitting and any stochastic preprocessing, together with known nondeterministic components. Artifact availability: access instructions for the cleaned dataset where redistribution is permitted, or a deterministic script when redistribution is restricted. License record: the source license text and a statement separating the Hugging Face repository license from any unverified upstream rights.

The final unique candidate count is particularly important. It determines the size of the ranking problem and the number of negatives considered for every query. It should be generated automatically from the exact final file rather than copied from an earlier experiment log. Data access and licensing

The immediate source is publicly accessible through Hugging Face and is labeled Apache-2.0. However, the repository's empty dataset card does not provide an upstream license analysis, attribution request, citation recommendation, or explanation of how the synonym pairs were obtained.

For this reason, the thesis should use measured language. It can state that the downloaded repository is labeled Apache-2.0, but it should not claim that the original terminology collection was independently verified as Apache-licensed unless an upstream source is identified. The repository identifier, access date, and pinned revision should be reported in the final dataset description. Limitations, Ethics, and Required Corrections Data-quality limitations

The most significant limitation is incomplete provenance. The source does not explain the collection population, annotation process, annotator qualifications, validation protocol, or intended benchmark task. Consequently, it is not possible to determine whether every pair represents strict synonymy, broader semantic relatedness, translation-mediated equivalence, or a mixture of these relations.

The observed examples and error analyses indicate that some labels are semantically debatable. For instance, a predicted term may be a plausible near-synonym but differ from the single recorded gold label. The dataset also includes relations involving abbreviations, register shifts, named entities, broader concepts, and domain-specific paraphrases. Therefore, exact-match failure cannot always be interpreted as a semantically incorrect system output.

The dataset provides one relevant answer per row, even when several synonyms may be valid. The ten duplicated inputs partially expose this issue but do not solve it, because the rows are still evaluated independently. This annotation incompleteness may underestimate system quality and may also influence comparisons between methods that prefer general synonyms and methods that prefer terminology closer to the dataset's wording.

A further limitation concerns representativeness. The source includes academic and domain terms, but no domain taxonomy or sampling design is documented. It is therefore not possible to claim that the dataset represents academic terminology generally, that disciplines are proportionally sampled, or that results generalize equally across fields. Ethical and privacy considerations

The retained project fields contain terminology strings rather than names, contact details, behavioral records, or other direct personal identifiers. On the basis of the supplied schema, no person-level information is used as model input. However, a formal personally identifiable information audit is not documented, and the source's upstream collection process remains unclear. Therefore, the absence of privacy risk should be described as an assessment of the retained fields rather than as a verified property of the complete upstream dataset.

The selected system sends the input term and its candidate shortlist to an external model

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through OpenRouter during reranking. Although the current data is not expected to contain personal information, this transfer creates an external-processing dependency. The exact provider, retention settings, regional processing conditions, and contractual privacy controls used for the experiments are unspecified in the data documentation. This would become a more significant concern if the pipeline were later applied to private institutional terminology, unpublished research topics, employee expertise profiles, or user queries.

Terminology resources can also encode cultural, disciplinary, and temporal assumptions. A synonym accepted within one academic community may be inappropriate or overly broad in another. Similarly, historically collected terminology can contain outdated naming conventions. Because the source lacks annotation dates and governance information, such risks cannot be quantified from the available materials.

The following safeguards are therefore appropriate:

only non-sensitive terminology should be sent to external APIs unless an approved data-processing arrangement exists; API request logging and provider retention settings should be documented; outputs should be described as ranked candidates rather than authoritative terminological equivalences; dataset errors and contested synonym relations should be preserved in an audit log rather than silently corrected; any manual corrections should be versioned and distinguished from the original source; future deployment in a specific academic domain should include validation by relevant subject-matter experts.

Required corrections before final submission

The current material is sufficient to explain the overall data design, but several items should be resolved before the chapter is treated as final.

First, the complete source-to-final filtering procedure must be documented. Relative to the currently published repository, 297 rows are absent from the final evaluation table, while only one exclusion is explicitly explained. A row-count audit should be generated directly from the preparation script.

Second, the final candidate-vocabulary cardinality must be computed and reported. The earlier figure of 921 unique values belongs to an earlier experimental state and cannot safely be assumed to describe the final 969-row dataset.

Third, canonical string handling must be specified. This includes whitespace trimming, casing, Unicode normalization, null handling, placeholder removal, punctuation, deduplication, and equality checks used for self-exclusion.

Fourth, the source revision must be pinned. Without a Hugging Face revision or source-file checksum, a future change to the public repository could produce different row counts and splits even when seed 42 is reused.

Fifth, the term-type classifier must be defined. The categories provide useful analytical evidence, but their tokenization rules and precedence are currently unspecified.

Finally, the task must consistently be identified as closed-pool, transductive synonym ranking for academic and domain terminology. This formulation accurately reflects the availability of the global candidate inventory, the hidden row-level mapping, and the limits of the conclusions that can be drawn from the reported metrics.

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